



Personalized stability monitoring based on body postures of construction workers working at heights

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ARTICLE INFO

Keywords:

Construction workers
Stability monitoring
Posture recognition
Working at height
Gaussian mixture model

ABSTRACT

Fall from heights accidents are one of the most frequent causes of death in the construction industry. Many fall-from-height accidents are related to stability issues and stability is critical for construction workers working at heights. Thus, stability monitoring of workers is important for proactive accident prevention. However, workers' stability is highly personalized due to differences in workers' physical characteristics and habits. Little attention has been paid to the personalized posture-based stability analysis of workers working at heights. This study proposes a personalized stability monitoring framework based on workers' body posture patterns when working at heights. The proposed method includes two main components: posture recognition and stability monitoring. First, OpenPose is used to extract the coordinates of workers' posture key points from video clips. Two posture features, duration and count, are extracted to reflect workload and frequency. Secondly, a two-stage stability monitoring framework, including instability detection and instability evaluation, is designed based on the Gaussian model and Gaussian mixture model. Finally, the proposed framework is validated by on-site construction videos of workers working at height. The validation results showed the accuracy of 84.38% and the precision of 81.25% in identifying the stable status of a subject. The robustness of the personalised monitoring method was validated through comparisons with five other workers. The study provides a practical reference for active safety monitoring for workers with high fall-from-height risk. It also helps to extend personalized and adaptive behaviour-based safety training for construction workers.

1. Introduction

The construction industry is a high-risk industry, and about 80–90 % of accidents are due to human-related causes (Salminen and Tallberg, 1996; Zhang and Fang, 2013). Falling from height is the leading cause of fatal injury (Chi and Han, 2013). Although measures have been developed to prevent serious fall from height accidents, e.g., mandatory personal safety protection equipment, they still occur frequently. Studies show that a common cause of these accidents is ineffective monitoring of workers' behaviours (Xia et al., 2018). Each construction worker's safety performance is closely related to the workers' individualised behavioural patterns, which are influenced by the different types of work, habits, experience, and other characteristics (Duan et al., 2022a). However, current safety monitoring systems do not consider the behavioural patterns of different workers, and they typically use a fixed safety threshold for all workers which result in frequent activation of false alarms (Chen et al., 2019; Guo et al., 2016; Yan et al., 2018). In

addition, current approaches do not detect abnormal events until after they have occurred, which is not conducive to proactive safety management (Chen et al., 2019). Thus, there is a need for further research on personalized safety monitoring methods.

Workers working at heights always need to be in a stable position to prevent serious fall-from-height accidents. Changes in the posture of workers during work will bring about changes in the individual's centre of gravity and stability. Once stability is lost, the risk of accidents will increase significantly. Moreover, workers' postural stability at height is highly personalized due to differences in workers' physical characteristics and habits (Cyma et al., 2018). Thus, a personalized stability monitoring method for workers working at heights is important to facilitate proactive accident prevention. However, there is a lack of attention to personalized stability analysis of workers working at height. In the medical field, to maximize the treatment effects and minimize side effects, the personalized medical theory suggests tailoring treatment measures according to each patient's predicted response or disease risks

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<https://doi.org/10.1016/j.ssci.2023.106104>

Received 22 June 2022; Received in revised form 22 December 2022; Accepted 13 February 2023

Available online 20 February 2023

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(Snyderman, 2012). Accordingly, statistical features of workers' posture patterns can be used to represent the workers' normal or acceptable status, and posture patterns that deviate from the norm can trigger alerts to supervisors (Kim et al., 2017; Yan et al., 2018).

Drawing inspiration from personalized medical theory, this study proposes a novel approach for monitoring workers' stability when working at heights. The basic concept is that when the observed stability-related action pattern of a worker deviates from the expected personalized action pattern, it can be regarded as a leading indicator of fall from height. The proposed approach aims to uncover latent safety information of construction workers working at height through automatic monitoring of observable action sequences. More specifically, the purpose of this paper is to propose a robust personalized stability monitoring framework based on the body postures of construction workers working at heights. To develop and illustrate the approach, this study focuses on construction workers installing external wall insulation. This is the first study to develop a personalized posture-based stability monitoring approach for construction workers working at heights. The proposed method can also serve as a reference for developing personalized safety monitoring approaches for other types of work.

The paper is organized as follows: Section 2 reviews past literature on posture-based safety and personalized construction safety monitoring approaches. Section 3 elaborates on the OpenPose algorithm, which is used to extract pose-related features. In combination with the Gaussian mixture model, features of multiple postures are integrated to establish a more robust personalized stability monitoring model. Section 4 describes how construction site videos are collected and used to validate the proposed framework. Lastly, Section 5 summarizes the main results and provides some limitations of this study for future research.

2. Literature review

2.1. Posture-related research in construction

The construction industry is still heavily dependent on manual labour. Construction workers' work postures are a valuable source of safety and productivity information that helps managers better control the construction process (Xu et al., 2022). Scholars have used posture information in different research areas, such as productivity analysis (Khosrowpour et al., 2014), workload analysis (Yu et al., 2019), and safety management (Han and Lee, 2013). There are many approaches for studying construction workers' postures. For example, Alwasel et al. (2017) used inertial measurement units (IMUs) and cameras to collect the kinematic data of masons, which were analysed using a support vector machine algorithm to classify the working postures of expert and non-expert masons. This produced a training system to improve the health and productivity of apprentice masons. In addition, Roberts et al. (2020) proposed a novel activity analysis framework for ergonomics analysis of individual construction workers based on deep learning and computer vision. The analysis framework estimates workers' two-dimensional posture and predicts workers' activities. Furthermore, Yu et al. (2019) proposed an automatic ergonomic load calculation model and established a detailed and accurate evaluation method based on biomechanical analysis.

Information about whether a construction worker is behaving safely can be obtained by observing changes in their posture. For instance, Han and Lee (2013) identified unsafe behaviour by comparing workers' behaviours and unsafe behaviour templates. In this study, using the example of ladder climbing, the authors use computer vision to detect the posture of workers and calculate their similarity to pre-collected posture templates to obtain safety information (Han and Lee, 2013). In addition, MassirisFernández et al. (2020) determined workers' ergonomic risk by comparing body posture angles with standard norms. By calculating Rapid Upper Limb Assessment scores from angles detected by computer vision, an automatic and accurate risk assessment can

be achieved (MassirisFernández et al., 2020). Thus, the first step in using body postures in safety monitoring is to specify and classify the work actions for monitoring. According to Presti and Cascia (2016), action classification methods can be categorised into: joint-based representations, mined joint-based descriptors, dynamics-based descriptors, similarity-based descriptors, and computer vision-based descriptors. Based on images, construction safety knowledge, and ergonomic principles, Guo et al. (2018) simplified dynamic motion into a static posture to identify unsafe behaviour and developed a skeleton-based real-time unsafe behaviour recognition method. In addition, Han et al. (2014) established a set of unsafe motion templates. They then used the similarity between the templates and the 3D human skeleton motion model extracted from videos of workers' motions to identify unstable movements on the ladders. Their study used motion data and similarity to identify the unstable status of workers. Similarly, Chen et al. (2019) determined the safety risk of workers' ladder climbing through similarity of worker's posture to typical postures given by experts. In addition, Chen et al. (2021) proposed a worker posture coding scheme, which can directly transform workers' body parts into a normalized 3D skeleton for safety monitoring. Combined with safety regulations and normalized skeleton, different posture targets can be described for safety behaviour monitoring. Table 1 summarises research in the construction industry that has previously applied action recognition methods and their results.

Previous studies observed unsafe behaviour based on changes in posture. However, the review shows that there is a lack of research focusing on posture-based stability analysis. Since stability is a key factor in fall-from-height accidents, the following section will review stability monitoring research in the construction industry.

2.2. Stability monitoring research in construction

On-site monitoring is a crucial safety management process to ensure construction safety. Construction safety monitoring still depends on human inspection and observation, which is labour-intensive, costly, inefficient, and error-prone (Awolusi et al., 2018; Chian et al., 2022; Goh et al., 2022). To ensure effective safety monitoring, the project manager has to formulate cumbersome monitoring processes and enforce strict monitoring standards. Due to the resources needed, individualized or personalized monitoring used to be infeasible. However, with the increasing adoption of technology in the construction industry, the use of computer vision, sensors, and building information models to facilitate detailed construction safety monitoring has gradually become attainable (Awolusi et al., 2018; Duan et al., 2022b). Two streams of research that tap on this monitoring data are reviewed: personalised safety management methods and stability monitoring of construction workers.

Personalised safety management methods have received research attention. For example, Tang et al. (2022) developed a personalized human factors analysis and classification system (HFACS) to monitor workers' real-time or near real-time status. The study defined personalized management indicators and data collection methods for unsafe behaviours. Guo et al. (2019) proposed a content-based and collaborative filtering-based method for recommending personalized safety training content. In addition, Yan et al. (2018) developed a worker-centred torso posture analysis system from the perspective of personalized medicine.

Due to the strong correlation between body movement patterns and incidents of slips, trips, and falls and (Yang and Ahn, 2019), stability monitoring has attracted significant attention in recent years. For example, Yang et al. (2017) proposed a collective perception method, which can perceive and evaluate the gait stability abnormalities of workers to identify the risk of falling at construction sites. Furthermore, Yang and Ahn (2019) proposed that the stability of workers can be analysed without knowing the threshold for identifying abnormality by investigating the spatial correlation of abnormal gait of workers. In

Table 1

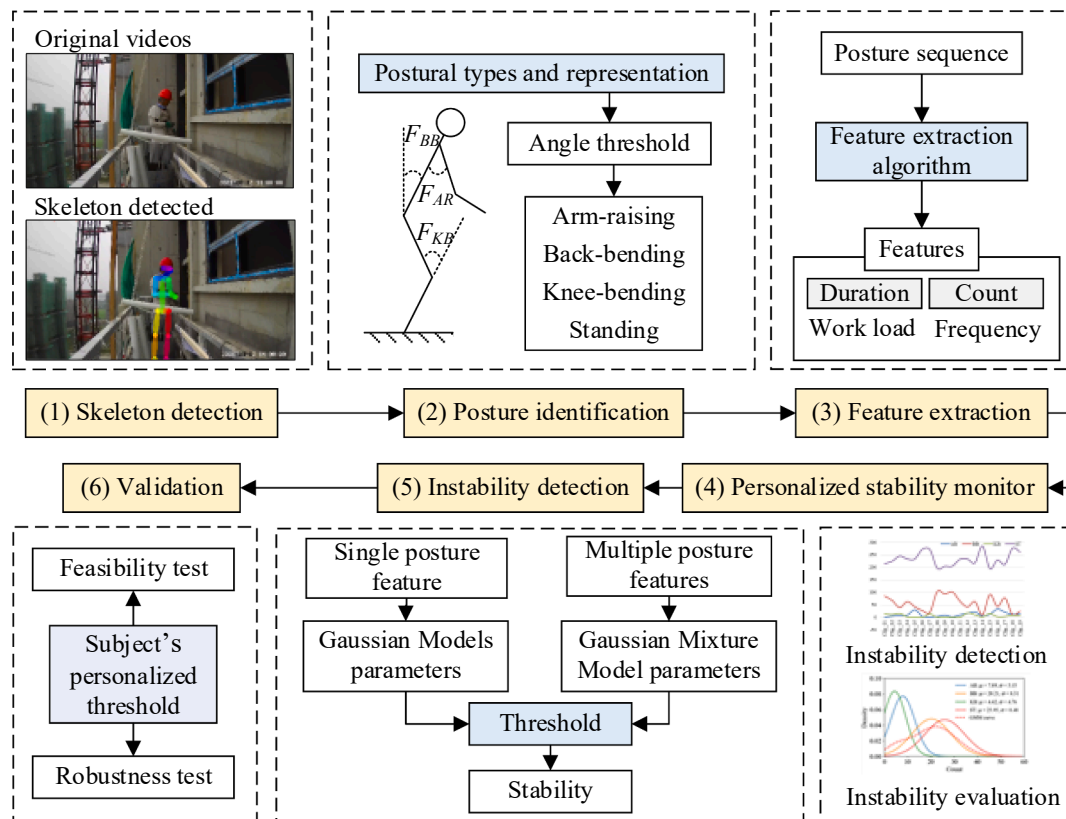
Previous studies of workers' action recognition for construction health and safety.

Action class	Data collection method	Classification method	Performance	Article
Standing, squatting, sitting, stooping, bending, and crawling	Computer vision-OpenNI	Linear discriminant analysis	Dataset-1 error rate: 5.2 %, dataset-2 error rate: 19.69 %	(Ray and Teizer, 2012)
Back-bending, arm-raising, and knee-bending	Computer vision - morphological variations	Support Vector Machine	Overall accuracy: 78.4 % – 88.6 %	(Seo and Lee, 2021)
Working overhead, kneeling, back bending forward, squatting, neck bending, and reaching	IMU-based motion capturing system	Support Vector Machine	Overall accuracy: 94.01 % ~100 %	(Chen et al., 2017)
Overhead working, squatting, stooping, semi-squatting, and one-legged kneeling	Wearable insole pressure system	Supervised machine learning classifiers	Overall accuracy: over 99 %	(Antwi-Afari et al., 2018)
Static bending, kneel on one leg and both legs, literal bend, transitional movements between postures, overhead work with one arm or both arms, squatting, standing, and without the knee touching ground	Low-cost wearable IMUs	Convolutional Neural Networks with Long Short-Term Memory (LSTM) layers,	Macro F1 Score: 0.870	(Zhao and Obonyo, 2020)
Gearing up, hammering, masonry, painting, roughcasting, sawing, screwing, sitting, standing still, and walking	Wearable MEMS-IMUs	Supervised machine learning classifiers	Average balanced accuracy: 85.54 % – 94.66 %	(Sanhudo et al., 2021)
Standing, bending-up, bending, bending-down, squatting-up, squatting, squatting-down, walking, twisting, working overhead, kneeling-up, kneeling, kneeling-down, and using stairs	Wearable IMUs	LSTM networks	Overall accuracy: 95.93 % – 97.60 %	(Kim and Cho, 2021)

another study, Umer et al. (2018) developed a static balance monitoring tool, which uses a wearable IMU and smartphone to actively track and monitor on-site construction personnel. Yoon et al. (2022) found that workers' gait significantly impacts their physical stability. Finally, Khan et al. (2021) proposed a correlation-based method to monitor unsafe behaviours of mobile scaffolds and construction workers working at heights.

Although recent studies have developed stability monitoring

approaches, there is still a lack of research on personalized stability management based on individualized stability patterns. The main reasons are the difficulties in collecting and managing personalized data due to its complex and dynamic nature (Tang et al., 2022), the large number of construction workers, the variety of work, and the heavy workload (Jeelani et al., 2021). Therefore, this paper proposes to measure the stability of different working at height workers based on their historical posture characteristics. This will help to improve personalized

**Fig. 1.** Research framework.

safety management of construction workers working at height.

3. Methodology

3.1. Research framework

Fig. 1 illustrates the framework for monitoring the stability of construction workers working at height. The framework includes five modules: skeleton detection, posture identification, feature extraction, personalized stability monitoring, instability detection, and validation. Skeleton detection is achieved through computer vision techniques. Compared to other skeleton extraction techniques (e.g., wearable sensors), this method has the advantage of being cost-effective and does not cause additional disruption to operations at height (Chian et al., 2021; MassirisFernández et al., 2020). Firstly, workers' skeleton and relative joint coordinates are detected based on images extracted from video frames. Postures that can reflect the stability of workers and behaviours are identified based on joint-related information. The time and duration of the posture are recorded based on the real-time feature extraction algorithm. Secondly, the collected features are imported into the personalized stability monitoring module and fed into a Gaussian model (GM) and a Gaussian mixture model (GMM) to calculate workers' personalized threshold for two-stage instability monitoring of stability detection and evaluation. Finally, the stability of worker is obtained by comparing his pose sequence features with the thresholds set. Other workers' pose sequence features are collected and compared with the initial worker's personalised threshold to analyse the robustness of the personalized method. Details of the modules, processes, and algorithms are illustrated in the following sections.

3.2. Skeleton detection

As the fundamental component of human behaviour, accurately detecting postures from video sequences is crucial for stability monitoring. This study uses OpenPose (Cao et al., 2021) to detect workers' postures and record body joint information. OpenPose is a bottom-up approach (the body nodes are detected first, and then the limbs are formed) for detecting human posture from digital images and videos. OpenPose allows real-time extraction of joint position. Compared with other human posture detection methods, OpenPose is superior in accuracy (Cao et al., 2021). It has shown promising results in construction research, such as ergonomic risk assessment (MassirisFernández et al., 2020) and safety helmet wearing detection (Gu et al., 2021). The algorithmic structure of OpenPose can be described as follows: first, primary features of human posture are extracted from the video frame. Two-branched parallel CNN are then used to predict different body parts' confidence maps and affinity fields. Finally, the part association and the whole skeleton of the person are generated based on bipartite matching from graph theory (Cao et al., 2021). The OpenPose algorithm enables users to extract 18, 25 or 70 joint positions from images, including facial and finger contours (Cao et al., 2021). Among these, the 25-joint model has advantages in terms of recognition speed and efficiency compared to other models, and has been widely used in construction-related research (Li et al., 2022; MassirisFernández et al., 2020). Therefore, the 25-joints model is selected in this study. The joint numbers and names are depicted in Fig. 2. The joint data extracted from videos will then be transferred to the second and third module as the input for posture recognition.

3.3. Posture identification

Postures, a fundamental element of human behaviour, can be considered a critical predictive feature of behaviours (Chen et al., 2021). For example, workers on a platform face a high risk of falling when they awkwardly extend their bodies over the platform. These awkward postures can provide valuable information for safety training, early

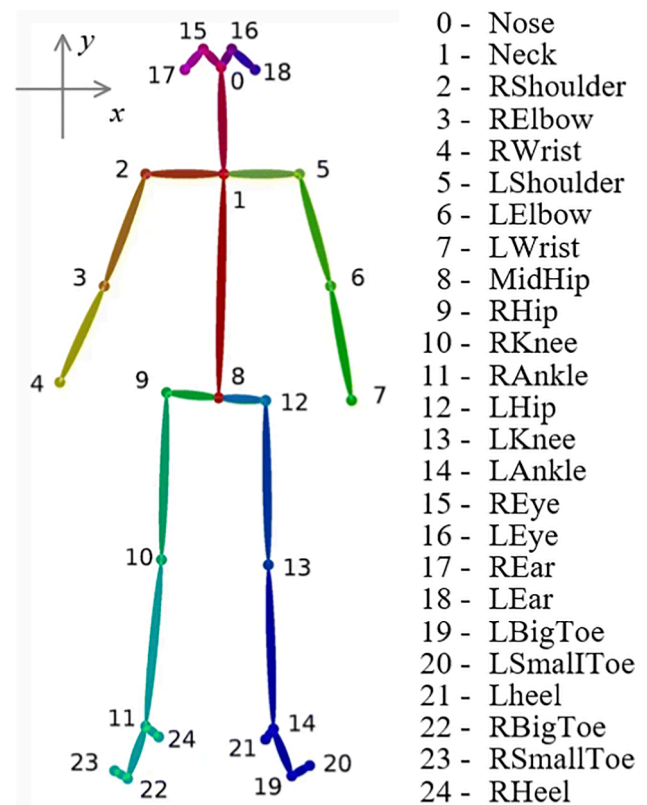


Fig. 2. Joints number and name in OpenPose.

warning, and management (Chen et al., 2021). Construction workers' safety risks can be recognized from some leading postures (Yu et al., 2017), which affect workers' stability and safety, regardless of trade and age (DiDomenico et al., 2010; Umer et al., 2018). For example, falling from height is a common accident type in construction. Many fall from height accidents occur when a worker becomes unbalanced during changes in the worker's centre of gravity. Therefore, when workers adjust their postures frequently, the likelihood of falling from height increases. Meanwhile, the leading postures should involve a considerable movement of human body parts and be repetitive to facilitate identification (Sherafat et al., 2020).

Table 1 lists leading postures that are indicative of postural stability when working at height. The main postures in construction include standing, climbing, bending, squatting, kneeling, sitting, and lying. In this study, three unstable postures that have a significant relationship with changes in the centre of gravity are selected as leading postures. The three unstable postures include arm-raising (AR), back-bending (BB), and knee-bending (KB). In addition, standing (ST) is listed as the posture representing the normal state. Referring to the practical parameter used in ergonomics risk assessment (MassirisFernández et al., 2020; Yan et al., 2017), Table 2 shows the posture representation method. Angles between different joint pairs are calculated by using the Cosine Rule.

3.4. Feature extraction

OpenPose extracts posture features of the construction workers based on leading postures and video frames. First, Fig. 3 illustrates the detailed workflow for recognising different postures. Frames extracted from video clips are used as inputs. The workflow outputs a vector of postures. It is noted that multiple postures may be detected in the same frame. If a posture is detected in the frame, the value corresponding to the output in the vector will be 1 for ST, 2 for AR, 3 for BB, and 4 for KB. If a posture is not detected, a value of 0 will be provided. However, based

Table 2
Leading postures and identification method.

Posture description	Angle name	Involved joint numbers	Formula	Threshold
Arm-raising (AR)	Left shoulder (SL)	∠03, 02, 09	$F_{AR} = \max(\text{SL}, \text{SR})$	$F_{AR} \in [90^\circ, 180^\circ]$
	Right shoulder (SR)	∠06, 05, 12		
Back-bending (BB)	Back-bend (BB)	∠01, 08, y-axis	$F_{BB} = \text{BB}$	$F_{BB} \in [20^\circ, 180^\circ]$
Knee-bending (KB)	Left Knee (KL)	∠09, 10, 11	$F_{KB} = 180^\circ - \min(\text{KL}, \text{KR})$	$F_{KB} \in [20^\circ, 180^\circ]$
	Right knee (KR)	∠12, 13, 14		
Standing (ST)	$F_{AR} \in [0^\circ, 90^\circ)$ AND $F_{BB} \in [0^\circ, 20^\circ)$ AND $F_{KB} \in [0^\circ, 20^\circ)$			

on the data collected, the leading postures in Table 2 last for about 0.5–1 s (about 8–15 frames), which can cause the values in the vector to vary across time. However, construction workers usually cannot change their posture so quickly. Thus, as a form of smoothing, the posture with the higher frequency across each set of 10 frames will be chosen as the representative posture for each set. In the example depicted in Fig. 4, a segment consisting of 10 frames is included, lasting approximately 0.7 s. The recognition results show that the worker's gestures in most of these frames are arm-raising. However, two of the frames are recognised as standing, with a standing time of only about 0.1 s. These are transient gestures and not representative of overall activity in the set. Therefore, the worker's poses in these 10 frames were classified based on the posture with the highest frequency, i.e., arm-raising. Eventually, the

workflow outputs the input video and a row vector composed of four numbers representing different postures of each frame. Then these vectors of different frames are combined to a postural matrix with a size of $1 \times 4n$ for feature extraction.

Subsequently, an algorithm (see Algorithm 1) extracts the count and duration of postures, which are used to indicate the stability of the worker. The algorithm calculates the duration and number of leading postures by recording the start and end time of the pose clips in the video frame. A higher frequency or duration of leading postures in the same period indicates that the worker is more likely to be unstable.

Algorithm 1 Pseudo code of postural feature extraction (For arm-raising)

```

Input: frame.action, Frames per second (FPS)
Output: duration, count
for each  $i = 1$  to length(frame.action) do
  if frame.action[i] == 2 & frame.action[i-1] != 2 then
    start =  $i * (1/\text{FPS})$ 
    count = count + 1
  else if frame.action[i] == 2 & frame.action[i-1] == 2 then
    end =  $i * (1/\text{FPS})$ 
    duration = end - start
  end if
end for
return duration, count

```

3.5. Personalized stability monitoring

The overall process of the personalized stability monitoring is illustrated in Fig. 5. The stability monitoring process is divided into two stages: (1) instability detection and (2) instability evaluation. The

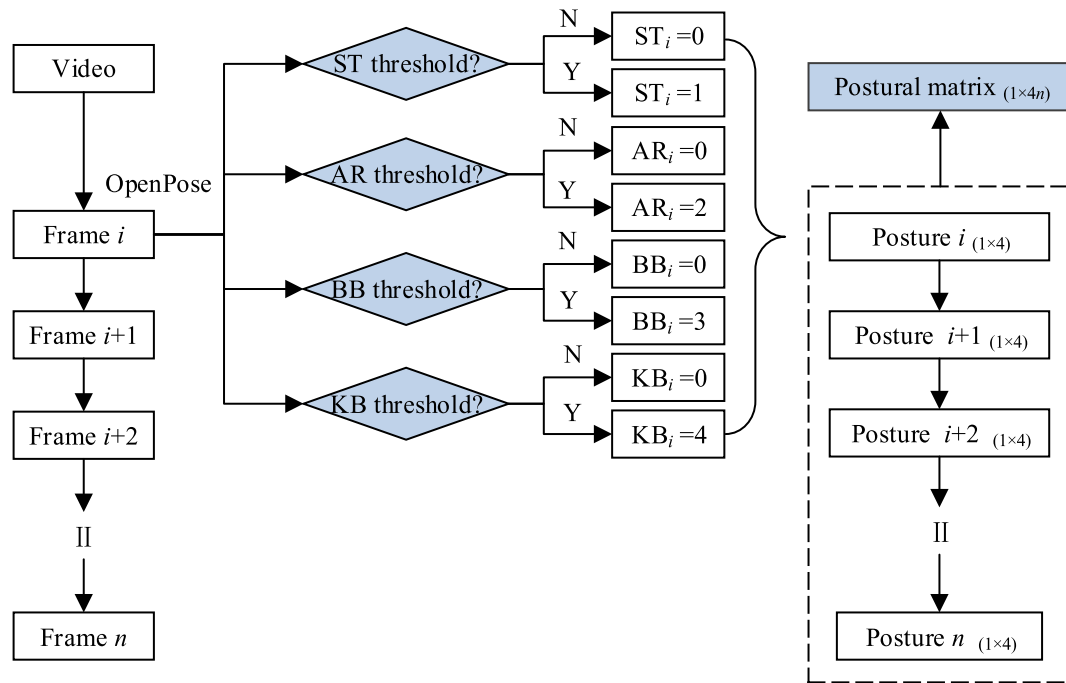


Fig. 3. Workflow for the recognition of different postures. AR: arm-raising; BB: back-bending; KB: knee-bending; ST: standing.

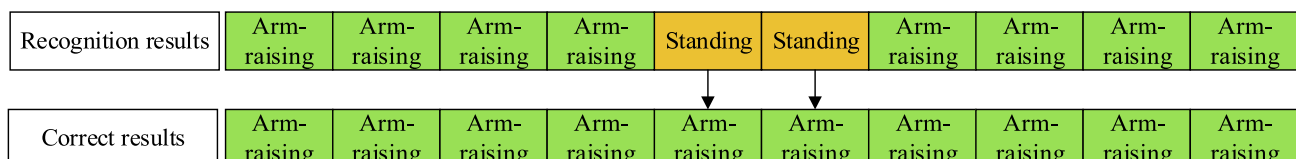


Fig. 4. Example for correction of recognition results.

former is meant to detect possible instances of instability, whereas the latter conducts a further evaluation to confirm the detected instability. An important characteristic of the proposed stability monitoring approach is the use of individual workers' historical data for parameter updating. During instability detection (stage 1), if a construction worker is detected to be in a stable state, the corresponding posture data will be stored in the safe posture sequence database to update personalized parameters used for future anomaly detection. Since there are 2 features used for judgment in stage 1, personalized parameters may frequently exceed the threshold. This means that even if the worker appears to be

unstable in terms of one feature, it is possible that the posture is actually stable. To enhance the robustness of the monitoring system and avoid misjudgement, the corresponding posture data will be input into stage 2 for further instability evaluation. If a worker is assessed to be unstable, which indicates exposure to a higher risk of falling, supervisors can then provide additional safety education or safety corrective measures for the worker together with the stored video frames.

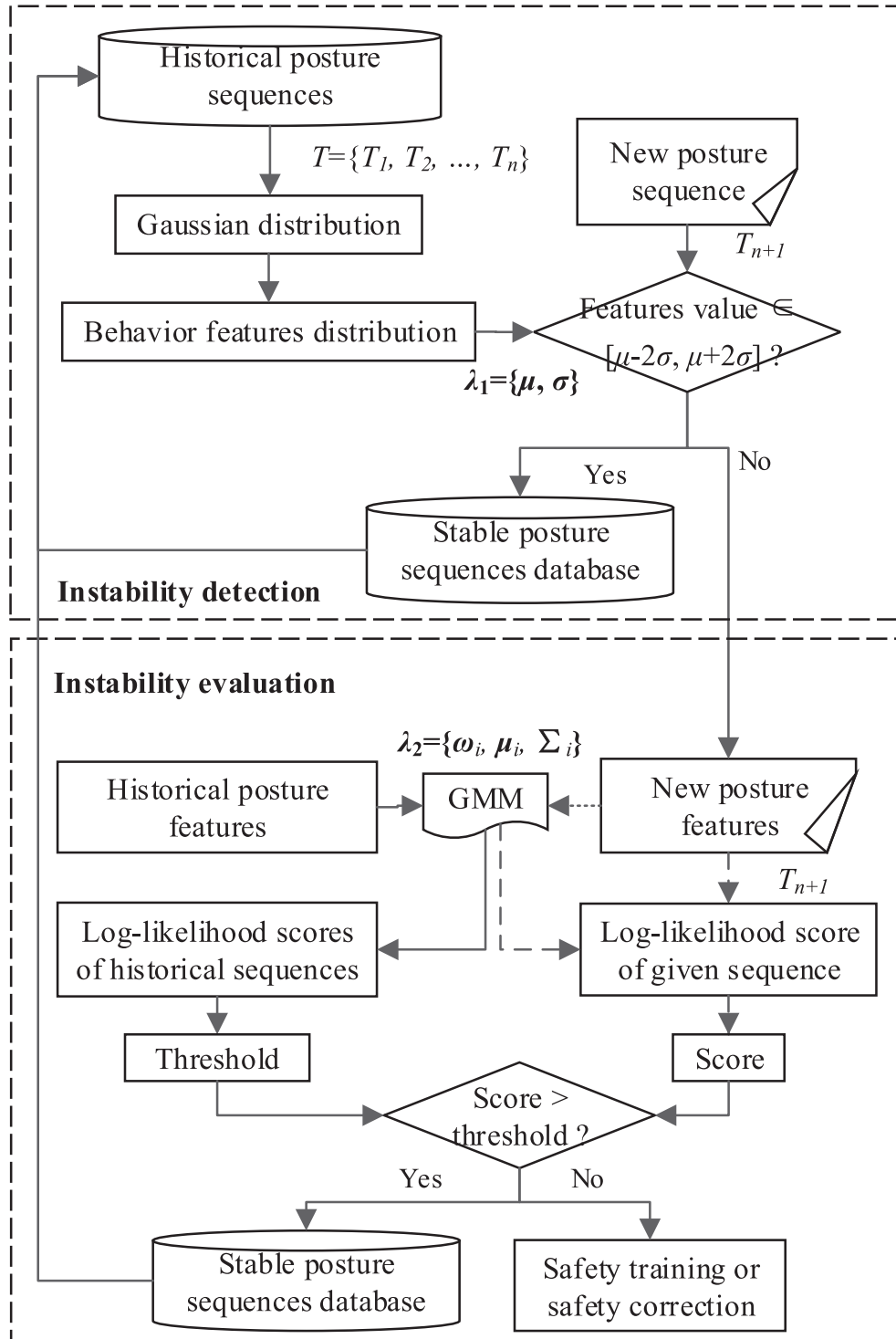


Fig. 5. Personalized stability monitoring.

3.6. Instability detection

According to personalized medical theory (Snyderman, 2012; Yan et al., 2018), the statistical average of a patient's behaviour reflects his or her acceptable status. Thus, the abnormal behaviours of workers can be detected and evaluated based on different statistics (Kim et al., 2017). Generally, basic features for daily behaviour obeys Gaussian-like probability distributions (GPD) (Meng et al., 2017). Hence, the Gaussian density function $f(x) \sim N(\mu, \sigma^2)$ is used in this study to describe expected patterns of each worker. The parameters of GPD includes $\lambda_1 = \{\mu, \sigma\}$, where μ refers to the average count and duration of different postures and σ is the standard deviation. These parameters are calculated based on the posture data extracted from the videos. The normal range for human behaviours is set as $P = [\mu - 2\sigma, \mu + 2\sigma]$ (Meng et al., 2017). If a worker's behaviour exceeds the upper limit, it is identified as a possible instance of instability. On the other hand, posture sequences that are within the interval $[\mu - 2\sigma, \mu + 2\sigma]$ is stored in the stable posture sequences database. Otherwise, it will be transferred to second stage of evaluation.

GMM is additionally used in this study to further evaluate if the detected instability is correctly detected (stage 2). GMM is an extension of the Gaussian model, which uses a combination of multiple Gaussian distributions to describe the data distribution. The parameter probability density function is expressed as the weighted sum of Gaussian component density, as shown in Eq. (1). Compared with other methods, GMM gives statistical inferences of each Gaussian distribution component and can be used to comprehensively analyse the anomaly posture sequence based on collected data (Li et al., 2016).

$$p(x|\lambda) = \sum_{i=1}^K \omega_i g(x|\mu_i, \sum_i) \quad (1)$$

In which, $\lambda_2 = \{\omega_i \mu_i, \sum_i\}$ are the parameter of GMM, K is the number of Gaussian distributions, ω is the mixed weight of multiple Gaussian distributions, and meets the constraint $\sum_{i=1}^K \omega_i = 1$, μ_i is the mean vector of component gaussian distribution $g(x|\mu_i, \sum_i)$, $\sum_i$ is the corresponding covariance matrix. For personalized stability monitor, we use the observed posture features as inputs and outputs workers' specific GMM parameters λ_2 for hidden risk state. The component Gaussian distribution number K is selected as 4. The most popular and mature expectation-maximization algorithm (EM) (Dempster et al., 1977) is adopted to calculate the model parameters λ_2 .

The model training input is expressed as X_{in} , including the features extracted from multiple monitoring segments. The log probabilities score was calculated for each monitoring segment. The stability monitoring threshold is set as the mean of all segment scores minus three times the standard deviation, as shown in Eq. (6) (Hosseinzadeh et al., 2006). When a new monitoring segment is inputted, the log-likelihood score of the segment is calculated and compared with the threshold. If the values are less than the threshold of corresponding features, it can be judged that the construction worker is unstable in the current segment, as shown in Algorithm 2. Otherwise, the clip of stable status will be recorded in the sequence database for personalized adjustment of posture features.

Algorithm 2 Pseudo code of personalized stability monitor of construction workers based on GMM

Input: Vectors of postural features, X_{in}
 Vectors of new postural features, X_{inn}
 The number of Gaussian components, K

Output: Status, GMM parameters λ_2

1. **Initialize** GMM parameters λ_{init}
2. **E step.** For each data sample, determine the a posteriori probability for each Gaussian component i .

$$P(i|x_{in}, \lambda_{init}) = \frac{\omega_i g(x_{in}|\mu_i, \sum_i)}{\sum_{j=1}^K \omega_j g(x_{in}|\mu_j, \sum_j)} \quad (2)$$

(continued on next column)

(continued)

Algorithm 2 Pseudo code of personalized stability monitor of construction workers based on GMM

3. **M step.** Update GMM parameters λ_2 . For each Gaussian component, update parameters using Eq. 3-5.

$$N = \text{length}(X_{in})$$

$$\omega_i^{new} = \frac{1}{N} \sum_{t=1}^N P(i|x_{in}, \lambda_{init}) \quad (3)$$

$$\mu_i^{new} = \frac{\sum_{t=1}^N P(i|x_{in}, \lambda_{init}) x_{in}}{\sum_{t=1}^N P(i|x_{in}, \lambda_{init})} \quad (4)$$

$$\sum_i^{new} = \frac{1}{\sum_{t=1}^N P(i|x_{in}, \lambda_{init})} \sum_{t=1}^N P(i|x_{in}, \lambda_{init}) (x_{in} - \mu_i^{new})(x_{in} - \mu_i^{new})^T \quad (5)$$

$$\lambda_2 = \{\omega_i^{new}, \mu_i^{new}, \sum_i^{new}\}$$

4. Set the likelihood threshold according to L , to detect unstable status.

$$L = \text{loglikelihood}(X_{in} | \lambda_2)$$

$$M = \text{length}(L)$$

$$\text{Threshold} = \frac{1}{M} \sum_{i=1}^M L_i - 3 \times \sqrt{\frac{1}{M} \sum_{i=1}^M \left(L_i - \frac{1}{M} \sum_{i=1}^M L_i \right)^2} \quad (6)$$

$$\text{Score} = \text{loglikelihood}(X_{inn}, \lambda_2)$$

if $\text{Score} < \text{Threshold}$ **then**

 Status = "Stable"

else

 Status = "Unstable"

end if

return Status, λ_2

4. Validation and results

Actual on-site construction activities are analysed to validate the proposed framework. The proposed method was implemented in the Python 3.7 environment, 64-bit Windows 10 system. With the PyTorch as a backend, those processes are accelerated based on an NVIDIA GeForce GTX 1080Ti GPU.

4.1. Data collection

The suspended scaffold is common construction equipment in which the hoisting mechanism is driven by the steel wire rope to lift the platform along the vertical surface during the high-rise building construction. It is mainly used to provide a movable platform for facade construction workers. Although the equipment has safety railings, there are still potential safety hazards associated with posture changes. First, during the construction of special-shaped exterior walls, it is necessary to lean over and even climb out of the suspended scaffold at height. Second, Frequent and large movements lead to the increase of platform swing amplitude, which may cause a breakdown of the platform.

A seven day data collection period on the construction site was carried out. The main work of all studied workers during the study was paving an external wall insulation board. To collect the key body nodes of construction workers as much as possible without interfering with the normal operation of the platform, the camera was fixed on the basket as shown in Fig. 6. Data collection was split: the first day for a pre-experiment test, four days of data collection for the training of the method, and two days for the validation. On all days of the experiment, data was captured from 8am to 5 pm. The video frame rate collected by the camera is 15 frames per second. By referring to previous research (Luo et al., 2020), five minute video slices were selected to form the data sets. Due to the weather, some video clips did not capture any workers. The final number of video clips in the data set for the subject A is 80. He is 36 years old and has more than 10 years of work experience.

The two-day validation was used to verify the feasibility and robustness of the personalized stability monitoring method described in this study. The video acquisition standards used were consistent to that used in the first five days. On the first day, five other clips were obtained of Subject A. On the second day, five new construction workers were monitored. Workers B and C were engaged in plastering the internal and external walls of the building, respectively. Worker D was engaged in tile paving. Workers E and F were responsible for installing the indoor



Fig. 6. Installation location of the camera.

ceiling. Five clips are extracted from each of the videos of the workers working. These workers belong to different construction trades and usually need to work at heights due to the nature of their work. The new construction workers' construction videos were obtained online due to the small number of workers working at height on the site. The data collection criteria and requirements for the online videos, and video properties (frame rate, video duration, etc.) were consistent with those collected in the field. The final data set includes 30 video clips.

4.2. Posture identification and feature extraction

A video clip was chosen from the proposed database to test the feature extraction performance. The postures of the construction worker in the video were annotated manually through a software when the video was viewed, the annotation result was determined as ground truth. The video annotation research tool ANVIL is used to determine the

posture sequence, as shown in Fig. 7. The actual count and duration of different postures can be calculated based on the sequence. Fig. 8(a)-(d) shows worker postural skeleton extraction results, as well as some typical examples of false recognition errors (Fig. 8(e)-(h)). The marked video length is 5 min, a total of 4500 frames. Comparing the results of the feature extraction with ground truth in Table 3, the performance for feature extraction was analysed. The result shows that the error of action features identified is within 5 %, which can meet the needs of stability monitoring. Subsequently, the construction video clips of construction workers were analysed to calculate personalized parameters.

4.3. Personalized stability monitoring

The actual stability of construction workers in the video was determined by human judgement. The judge is an expert with extensive experience in on-site safety management. Based on his knowledge, a

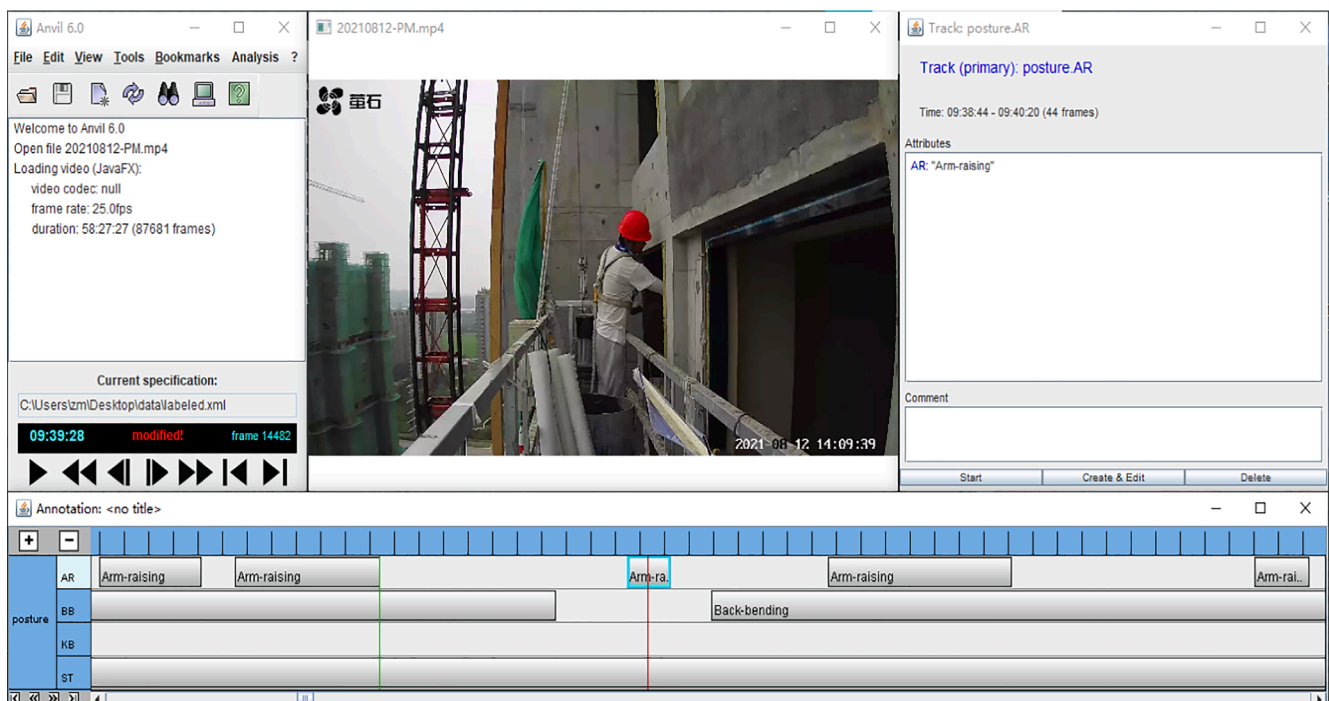


Fig. 7. User interface of ANVIL.

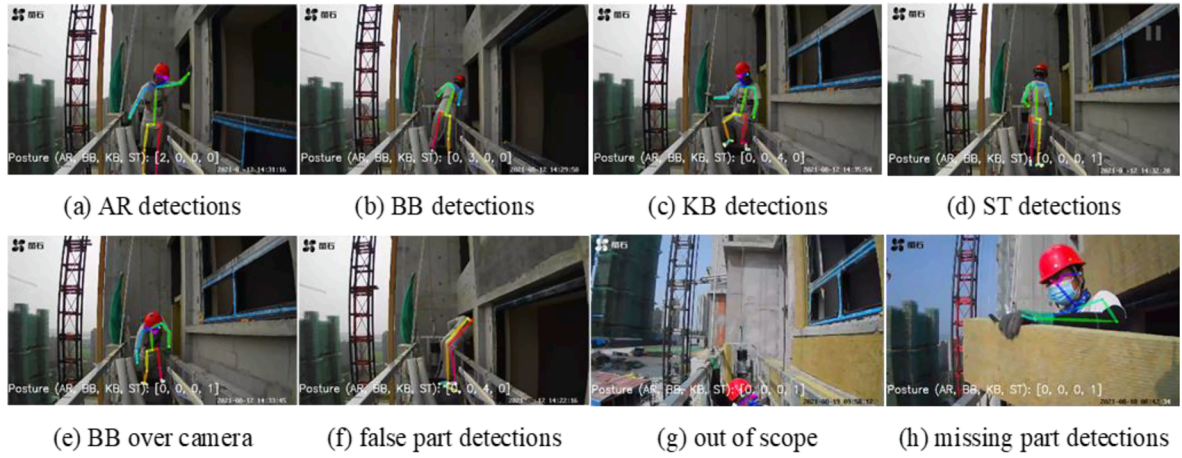


Fig. 8. Examples of posture identification results.

Table 3
Performance metrics of feature extraction.

	Duration/s		Count	
	Model	Truth	Model	Truth
Overall	312.00	308.13	82	79
AR	22.00	36.60	17	16
BB	62.67	76.00	20	16
KB	10.00	6.53	9	14
ST	217.33	189.00	36	33

worker was judged to be unstable if the video showed a lot of bending, leaning out, and other movements that could cause a shift in the centre of gravity. 48 videos judged to be stable were selected to calculate the personalized parameters of the subject A. The parameter $\lambda = \{\lambda_1, \lambda_2\} = \{\mu, \sigma, \omega_i, \mu_i, \Sigma_i\}$ ($i = 1, 2$) reflects the stability of workers as calculated based on the proposed method. The λ_1 is calculated based on GM, and the other one λ_2 is calculated by GMM.

Figs. 9-10 shows the fitted curve of based on worker's postural features and GM for calculating worker's personalized stability monitoring interval $[\mu - 2\sigma, \mu + 2\sigma]$. The reasonable duration within the 5-minute video slices of AR is [0 s, 38.08 s], and the reasonable number of occurrences is [0, 18.57]; the reasonable duration of BB is [0 s, 96.38 s], and the reasonable number of occurrences is [0.70, 31.14]; the reasonable duration of KB is [0 s, 25.87 s], and the reasonable number of occurrences is [0, 23.30]; the reasonable duration of ST is [192.43 s, 295.87 s], and the reasonable number of occurrences is [5.33, 43.21]. The worker was detected standing most of the time in the dataset, with a few instances knee-bending. On the suspended scaffold, the worker mainly works on the external wall of the building and must bend

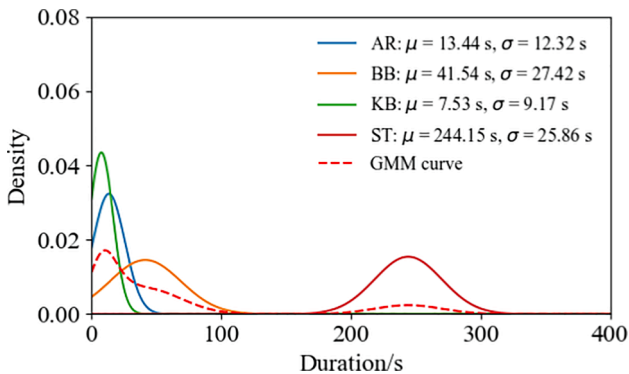


Fig. 9. The duration of different postures based on GM.

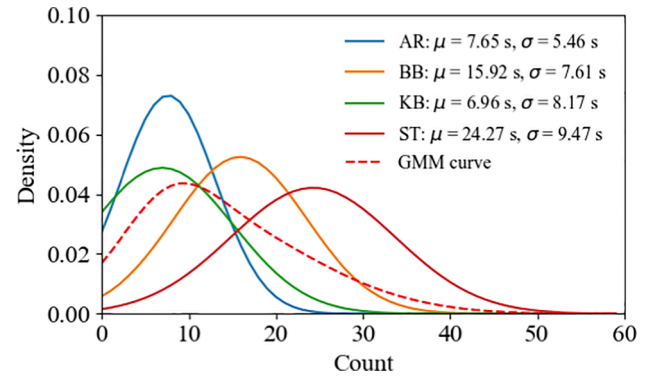


Fig. 10. The count of different postures based on GM.

outward during operation. Bending, particularly with the upper limbs bending beyond the area of the scaffold, poses a certain risk of falling from height.

The sklearn package in python is used to establish GMMs for extract posture features (duration and count), with the three parameters $\lambda_2 = \{\omega_i, \mu_i, \Sigma_i\}$ ($i = 1, 2$) being calculated based on the EM algorithm. Among them, the calculated weight of components are as follows: for the duration, the weight of component Gaussian distributions is $\omega_1 = \{0.2283, 0.2908, 0.3767, 0.1042\}$. For the number of occurrences, the weight of component Gaussian distributions is $\omega_2 = \{0.1439, 0.2659, 0.2918, 0.2983\}$. The related GMM curve is shown in Figs. 9-10. Different from instability detection module, the instability evaluation considers the change characteristics of various postures, which can avoid misjudgement and has higher robustness. The log-likelihood of each posture feature of each video clip is calculated in turn, and the threshold is calculated according to the method described in Section 3. The final threshold for the feature of duration is -17.68 , and that of count is -16.00 .

4.4. Instability detection and validation

A validation experiment was conducted to verify that the feasibility and robustness of stability detection method proposed in this study. The validation data set can be divided into two parts: the first part is the construction videos of the construction worker in the formal experiment for feasibility test, which is recorded as subject A. The second part is the videos of five other construction workers for robustness test, which are recorded as subject B, C, D, E, and F.

4.4.1. Feasibility test

The feasibility of the monitoring framework is validated based on videos of subject A. The number of validation videos is 32. Based on human judgement, there are 15 clips judged to be stable, and 17 clips that are unstable. The instability monitoring results are shown in Figs. 11–12. In Fig. 12, a column height of 1 means unstable and 2 means stable. First, according to the results of instability detection, most of the changes observed for posture features fall within reasonable ranges. However, the duration of BB and KB clearly exceeded the stable range. Taking the first 20 video clips in Fig. 12 as an example, the results of instability detection show that many clips are detected to be unstable. Second, the performance of the proposed instability monitoring method is evaluated with reference to the performance metrics in machine learning (Chian et al., 2022). This is because there is no unified standard for the classification performance of posture description currently. The main evaluation metrics include accuracy, precision, recall, and F1 score. The result shows that the accuracy for the verification videos is 84.38 %, the precision is 81.25 %, the F1 score is 86.67 %, and the recall is 83.87 %. These indicators show that the proposed stability monitoring framework can identify the stability of workers working at height. Moreover, project managers can analyse the main types of unstable postures of worker instability by combining the results of these two modules.

4.4.2. Robustness test

The robustness of the monitoring framework is validated by judging the stability of other workers using subject A's individual parameters. The robustness test includes including five new male construction workers other than the subject A. The movements of these workers in the clips are all stable according to human judgement. First, stability detection was conducted to analyse whether other workers' posture feature parameters are within the subject A's reasonable interval. The results are shown in Fig. 13. Their posture features at heights shows completely different characteristics and many of the features are outside the reasonable range of subject A. Due to the differences between individuals, the reasonableness of their stability cannot be determined by subject A's personalized parameters.

Second, the instability evaluation results of the construction workers as reflected by each feature parameter is described based on the GMM. Results are shown in Figs. 14–15. The red dotted line is the calculated instability evaluation threshold of subject A. Any scores lower than suggest that the subjects are unstable. Most of their scores are significantly lower than the threshold set from subject A. While the workers' performance in the video clips fully complies with safety regulations, the model suggests the opposite. This shows that it is unreasonable to use this threshold to monitor the stability of other workers. This verifies that

this calculated threshold is personalized to subject A.

5. Discussion

This paper proposes a personalized stability monitoring method based on body postures for construction workers working at height. It consists of five components: skeleton detection, posture representation, feature extraction, personalized stability monitoring, and instability detection. OpenPose is used to extract the workers' digital skeleton coordinates from the monitoring video. A leading posture is used to describe a variety of statuses of workers working at heights. Posture features of workers are further extracted from workers' posture sequence. The personalized stability parameters are identified in the stability monitoring module based on GM and GMM. The validation results show that identification accuracy for a particular subject is 84.38 % and the precision is 81.25 %. By comparing the personalised stability parameters with those of 5 other workers, the robustness of the personalised monitoring method is validated.

This study is the first to use computer vision to study the posture of construction workers working at height to determine their personalized stability. As a proactive monitoring method, this study has achieved desirable results in stability monitoring and doing so based on a personalized system. The method is based on surveillance video to judge worker stability, without relying on constant human judgement or expensive wearable devices. Currently, the method only considers the features of workers' postures. If this is combined with the characteristics of workers' different backgrounds, personalities, or habits, a more comprehensive individualized safety monitoring system for workers can be established. By formulating differentiated stability monitoring and correction programs, personalized safety training programs can be customized for different construction workers, thereby improving the level of active safety management in the construction industry. Ultimately, the method can be extended to other tasks that have high demands on postural stability by changing the monitored objects.

The contributions of this research are follows. Firstly, this study provided insight into the implementation of proactive site surveillance by introducing a posture-based personalized stability monitoring method. Based on previous studies (Yang and Ahn, 2019), unsafe behaviour of workers working at height is closely linked to changes in an individual's centre of gravity and stability. However, most personalized monitoring methods in existing studies use unified monitoring parameters for individualized monitoring of different workers. Little attention has been paid to the personalized stability monitoring based on construction workers differentiated expected postural patterns. Moreover, because of different types of work and different personality characteristics, stability-related information will be hidden in workers' normal

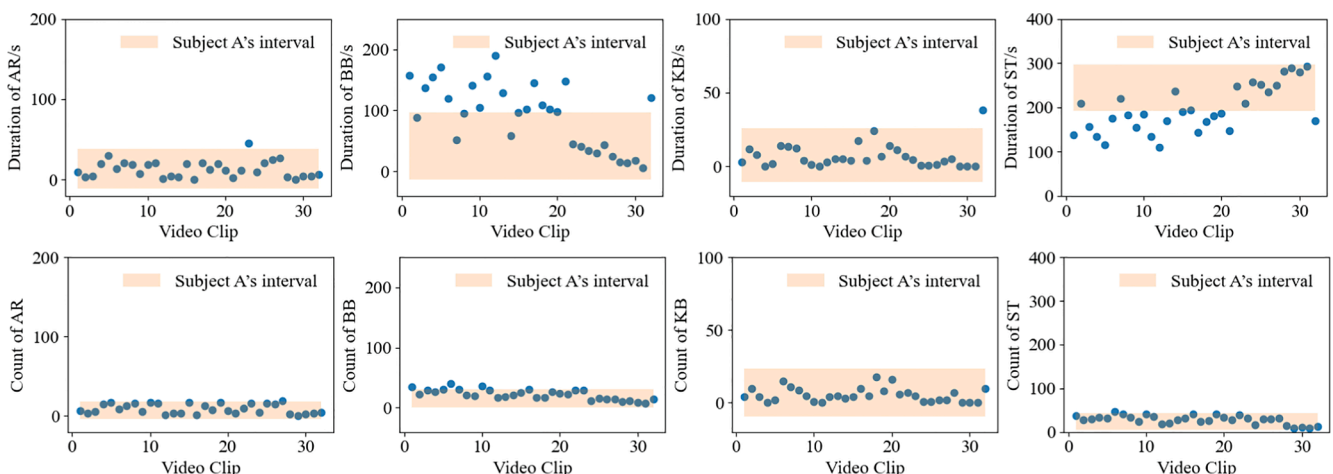


Fig. 11. Feasibility test results of instability detection.

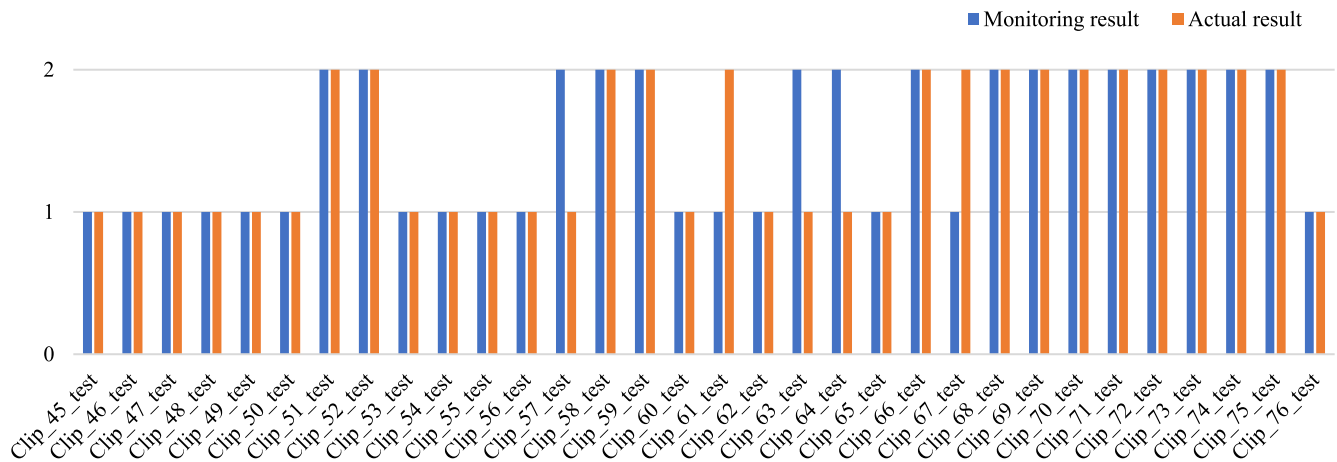


Fig. 12. Feasibility test results of instability evaluation.

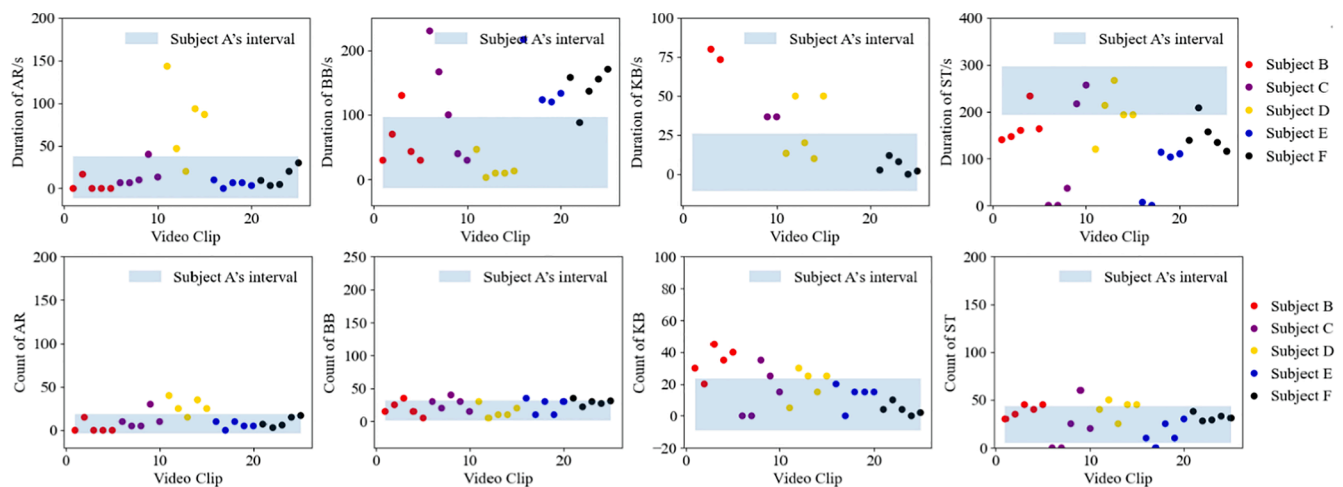


Fig. 13. Robustness test results of instability detection.

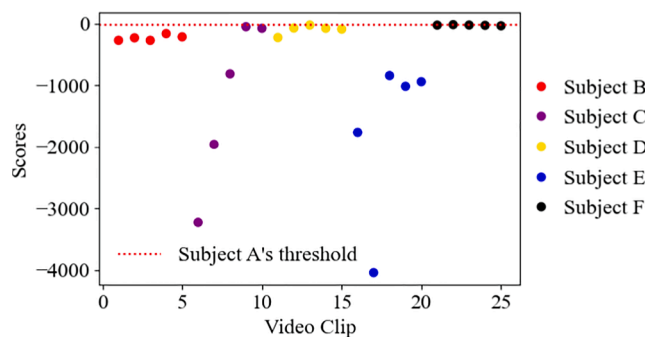


Fig. 14. Duration comparison of instability evaluation.

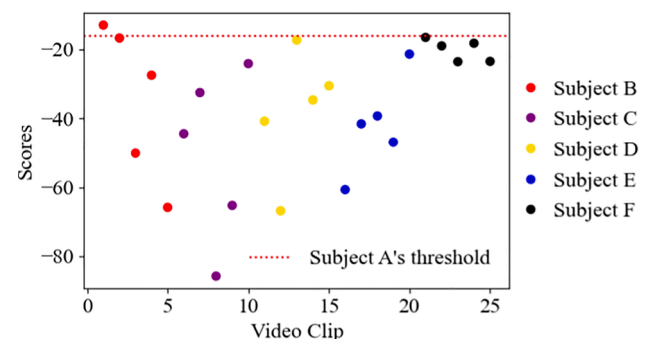


Fig. 15. Count comparison of instability evaluation.

operation postures. This study uses a personalized pattern to set a threshold for stability as demonstrated workers working at height. The feasibility and robustness of this method is demonstrated studies of workers on a construction site. Secondly, Gaussian mixture model is applied to establish a two-stage personalized stability monitoring framework for construction workers working at height. Compared to traditional safety monitoring systems in previous studies, the Gaussian mixture model is established by integrating multiple postural features for stability evaluation. The proposed method is not only helpful for managers to evaluate the instability of construction workers working at

height, but also can help managers to dig out the main types of unstable postures why construction workers are in the unstable status from the video. The method can also be utilized to obtain some insights into anomalous behaviours other than stability. For example, the fatigue of construction workers may lead to abnormal knee bending.

The proposed personalised stability monitoring method has practical implications for the safety of workers working at height. The study shows that the method can be easily implemented in construction projects to proactively identify workers with a tendency to be unstable. One of the advantages of the method is its ability to monitor workers without

interfering with their work or requiring workers to wear additional sensors. Furthermore, the accuracy of the system can be improved for each individual worker if managers regularly collect video clips of the workers working to update the personalized parameters. Abnormalities in workers' behavioural parameters can also be automatically monitored and managers can investigate proactively. A personalized or adaptive behaviour-based safety programme can be created for workers working at heights, where the observations and feedback are facilitated by the proposed method. Safety personnel can proactively coach and train workers who are prone to instability and have higher fall-from-heights risk.

6. Conclusion

Workers' stability is an important factor influencing the safety of construction workers working at heights. Many automatic after-the-event monitoring systems do not consider the behavioural patterns of different workers and give false alarms, which is not conducive to active safety management. This is due to workers displaying personalised patterns of stability, with different characteristics. This study presents a two-stage stability monitoring framework based on the posture features of construction workers working at height. Field experiments verify the feasibility and robustness of the proposed method. The proposed method has significant practical implications in terms of increasing active safety monitoring in the construction industry, assisting project stakeholders in taking the initiative to control the workers' safety. It also provides a foundation for formulating a reasonable behaviour-based safety management plan prior to possible construction safety accidents.

There are some limitations in this study that may be resolved by future research. Due to the specific tasks used for this study, the personalised stability monitoring focused predominantly on males. While the practical validation for different genders needs to be further studied. Key body node extraction based on OpenPose will be affected by light, angle, obstacles, and so on. Although the worker's skeleton can be completely extracted most of the time, during the on-site data collection, it was found that if too many construction materials are placed on the platform, the key points of knee and foot posture will be blocked. In addition, the key point coordinates of construction workers extracted in this paper are two-dimensional, which will lead to deviation in some posture judgments. For example, if the worker bends over the lens, it will be difficult to extract the action features in the video accurately. Therefore, more advanced algorithms can be considered to extract a three-dimensional skeleton of workers. To extend this method to other tasks, it would be worthwhile to further investigate how to continuously collect videos of workers working at height in other trades.

CRediT authorship contribution statement

Pinsheng Duan: Conceptualization, Methodology, Software, Validation, Writing- original draft. **Yang Miang Goh:** Methodology, Writing- review & editing. **Jianliang Zhou:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This study is funded by the National Natural Science Foundation of

China (Number: 72171224), the Humanities and Social Sciences Foundation of China's Education Ministry (Number: 19YJAZH122), Graduate Innovation Program of China University of Mining and Technology (Number: 2021WLKXJ033), and the Doctor Joint-Training Program of China Scholarship Council (Number: 202106420087)

References

- Alwasel, A., Sabet, A., Nahangi, M., Haas, C.T., Abdel-Rahman, E., 2017. Identifying poses of safe and productive masons using machine learning. *Autom. Constr.* 84, 345–355. <https://doi.org/10.1016/j.autcon.2017.09.022>.
- Antwi-Afari, M.F., Li, H., Yu, Y., Kong, L., 2018. Wearable insole pressure system for automated detection and classification of awkward working postures in construction workers. *Autom. Constr.* 96, 433–441. <https://doi.org/10.1016/j.autcon.2018.10.004>.
- Awolusi, I., Marks, E., Hallowell, M., 2018. Wearable technology for personalized construction safety monitoring and trending: review of applicable devices. *Autom. Constr.* 85, 96–106. <https://doi.org/10.1016/j.autcon.2017.10.010>.
- Cao, Z., Hidalgo, G., Simon, T., Wei, S.E., Sheikh, Y., 2021. OpenPose: realtime multi-person 2D pose estimation using part affinity fields. *IEEE Trans. Pattern Anal. Mach. Intell.* 43, 172–186. <https://doi.org/10.1109/TPAMI.2019.2929257>.
- Chen, H., Luo, X., Zheng, Z., Ke, J., 2019. A proactive workers' safety risk evaluation framework based on position and posture data fusion. *Autom. Constr.* 98, 275–288. <https://doi.org/10.1016/j.autcon.2018.11.026>.
- Chen, H., Luo, X., Zhu, Z., 2021. A worker posture coding scheme to link automatic and manual coding. *Autom. Constr.* 125, 103630. <https://doi.org/10.1016/j.autcon.2021.103630>.
- Chen, J., Qiu, J., Ahn, C., 2017. Construction worker's awkward posture recognition through supervised motion tensor decomposition. *Autom. Constr.* 77, 67–81. <https://doi.org/10.1016/j.autcon.2017.01.020>.
- Chi, S., Han, S., 2013. Analyses of systems theory for construction accident prevention with specific reference to OSHA accident reports. *Int. J. Proj. Manag.* 31, 1027–1041. <https://doi.org/10.1016/j.ijproman.2012.12.004>.
- Chian, E., Fang, W., Goh, Y.M., Tian, J., 2021. Computer vision approaches for detecting missing barricades. *Autom. Constr.* 131, 103862. <https://doi.org/10.1016/j.autcon.2021.103862>.
- Chian, E.Y.T., Goh, Y.M., Tian, J., Guo, B.H.W., 2022. Dynamic identification of crane load fall zone: a computer vision approach. *Saf. Sci.* 156, 105904. <https://doi.org/10.1016/j.ssci.2022.105904>.
- Cyma, M., Marciniak, K., Tomczak, M., Stemplewski, R., 2018. Postural stability and physical activity of workers working at height. *Am. J. Mens. Health* 12, 1068–1073. <https://doi.org/10.1177/1557988318774996>.
- Dempster, A.P., Laird, N.M., Rubin, D.B., 1977. Maximum likelihood from incomplete data via the EM algorithm. *J. R. Stat. Soc. Ser. B* 39, 1–22. <https://doi.org/10.1111/j.2517-6161.1977.tb01600.x>.
- DiDomenico, A., McGorry, R.W., Huang, Y.H., Blair, M.F., 2010. Perceptions of postural stability after transitioning to standing among construction workers. *Saf. Sci.* 48, 166–172. <https://doi.org/10.1016/j.ssci.2009.07.006>.
- Duan, P., Zhou, J., Fan, W., 2022a. Safety tag generation and training material recommendation for construction workers: a persona-based approach. *Eng. Constr. Archit. Manag.* <https://doi.org/10.1108/ECAM-12-2021-1143> ahead-of-p.
- Duan, P., Zhou, J., Tao, S., 2022b. Risk events recognition using smartphone and machine learning in construction workers' material handling tasks. *Eng. Constr. Archit. Manag.* ahead-of-p. doi: 10.1108/ECAM-10-2021-0937.
- Goh, Y.M., Tian, J., Chian, E.Y.T., 2022. Management of safe distancing on construction sites during COVID-19: a smart real-time monitoring system. *Comput. Ind. Eng.* 163, 107847. <https://doi.org/10.1016/j.cie.2021.107847>.
- Gu, Y., Wang, Y., Shi, L., Li, N., Zhuang, L., Xu, S., 2021. Automatic detection of safety helmet wearing based on head region location. *IET Image Process.* 15, 2441–2453. <https://doi.org/10.1049/ipr2.12231>.
- Guo, S.Y., Ding, L.Y., Luo, H.B., Jiang, X.Y., 2016. A Big-Data-based platform of workers' behavior: observations from the field. *Accid. Anal. Prev.* 93, 299–309. <https://doi.org/10.1016/j.aap.2015.09.024>.
- Guo, S., Ding, L., Zhang, Y., Skibniewski, M.J., Liang, K., 2019. Hybrid recommendation approach for behavior modification in the Chinese construction industry. *J. Constr. Eng. Manag.* 145, 04019035. [https://doi.org/10.1061/\(asce\)co.1943-7862.0001665](https://doi.org/10.1061/(asce)co.1943-7862.0001665).
- Guo, H., Yu, Y., Ding, Q., Skitmore, M., 2018. Image-and-skeleton-based parameterized approach to real-time identification of construction workers' unsafe behaviors. *J. Constr. Eng. Manag.* 144, 04018042. [https://doi.org/10.1061/\(asce\)co.1943-7862.0001497](https://doi.org/10.1061/(asce)co.1943-7862.0001497).
- Han, S., Lee, S., 2013. A vision-based motion capture and recognition framework for behavior-based safety management. *Autom. Constr.* 35, 131–141. <https://doi.org/10.1016/j.autcon.2013.05.001>.
- Han, S., Lee, S., Peña-Mora, F., 2014. Comparative study of motion features for similarity-based modeling and classification of unsafe actions in construction. *J. Comput. Civ. Eng.* 28, A4014005. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000339](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000339).
- Hosseinizadeh, D., Krishnan, S., Khademi, A., 2006. Keystroke identification based on Gaussian mixture models. In: ICASSP, 2006 IEEE International Conference on Acoustics Speech and Signal Processing Proceedings. pp. III–III. doi: 10.1109/icassp.2006.1660861.

- Jeelani, I., Asadi, K., Ramshankar, H., Han, K., Albert, A., 2021. Real-time vision-based worker localization & hazard detection for construction. *Autom. Constr.* 121, 103448 <https://doi.org/10.1016/j.autcon.2020.103448>.
- Khan, N., Saleem, M.R., Lee, D., Park, M.W., Park, C., 2021. Utilizing safety rule correlation for mobile scaffolds monitoring leveraging deep convolution neural networks. *Comput. Ind.* 129, 103448 <https://doi.org/10.1016/j.compind.2021.103448>.
- Khosrowpour, A., Niebles, J.C., Golparvar-Fard, M., 2014. Vision-based workplace assessment using depth images for activity analysis of interior construction operations. *Autom. Constr.* 48, 74–87. <https://doi.org/10.1016/j.autcon.2014.08.003>.
- Kim, H., Ahn, C.R., Yang, K., 2017. Identifying safety hazards using collective bodily responses of workers. *J. Constr. Eng. Manag.* 143, 04016090. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001220](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001220).
- Kim, K., Cho, Y.K., 2021. Automatic recognition of workers' motions in highway construction by using motion sensors and long short-term memory networks. *J. Constr. Eng. Manag.* 147, 04020184. [https://doi.org/10.1061/\(asce\)co.1943-7862.0002001](https://doi.org/10.1061/(asce)co.1943-7862.0002001).
- Li, L., Hansman, R.J., Palacios, R., Welsch, R., 2016. Anomaly detection via a Gaussian Mixture Model for flight operation and safety monitoring. *Transp. Res. Part C Emerg. Technol.* 64, 45–57. <https://doi.org/10.1016/j.trc.2016.01.007>.
- Li, J., Zhao, X., Zhou, G., Zhang, M., 2022. Standardized use inspection of workers' personal protective equipment based on deep learning. *Saf. Sci.* 150, 105689 <https://doi.org/10.1016/j.ssci.2022.105689>.
- Luo, H., Liu, J., Fang, W., Love, P.E.D., Yu, Q., Lu, Z., 2020. Real-time smart video surveillance to manage safety: a case study of a transport mega-project. *Adv. Eng. Informatics* 45, 101100. <https://doi.org/10.1016/j.aei.2020.101100>.
- Massiris-Fernández, M., Fernández, J.A., Bajo, J.M., Delrieux, C.A., 2020. Ergonomic risk assessment based on computer vision and machine learning. *Comput. Ind. Eng.* 149, 106816 <https://doi.org/10.1016/j.cie.2020.106816>.
- Meng, L., Miao, C., Leung, C., 2017. Towards online and personalized daily activity recognition, habit modeling, and anomaly detection for the solitary elderly through unobtrusive sensing. *Multimed. Tools Appl.* 76, 10779–10799. <https://doi.org/10.1007/s11042-016-3267-8>.
- Presti, L.L., Cascia, M.L., 2016. 3D skeleton-based human action classification: a survey. *Pattern Recognit.* 53, 130–147. <https://doi.org/10.1016/j.patcog.2015.11.019>.
- Ray, S.J., Teizer, J., 2012. Real-time construction worker posture analysis for ergonomics training. *Adv. Eng. Informatics* 26, 439–455. <https://doi.org/10.1016/j.aei.2012.02.011>.
- Roberts, D., Torres Calderon, W., Tang, S., Golparvar-Fard, M., 2020. Vision-based construction worker activity analysis informed by body posture. *J. Comput. Civ. Eng.* 34, 04020017. [https://doi.org/10.1061/\(asce\)cp.1943-5487.0000898](https://doi.org/10.1061/(asce)cp.1943-5487.0000898).
- Salminen, S., Tallberg, T., 1996. Human errors in fatal and serious occupational accidents in Finland. *Ergonomics* 39, 980–988. <https://doi.org/10.1080/00140139608964518>.
- Sanhudo, L., Calvetti, D., Martins, J.P., Ramos, N.M.M., Méda, P., Gonçalves, M.C., Sousa, H., 2021. Activity classification using accelerometers and machine learning for complex construction worker activities. *J. Build. Eng.* 35, 102001 <https://doi.org/10.1016/j.job.2020.102001>.
- Seo, J., Lee, S., 2021. Automated postural ergonomic risk assessment using vision-based posture classification. *Autom. Constr.* 128, 103725 <https://doi.org/10.1016/j.autcon.2021.103725>.
- Sherafat, B., Ahn, C.R., Akhavan, R., Behzadan, A.H., Golparvar-Fard, M., Kim, H., Lee, Y.-C., Rashidi, A., Azar, E.R., 2020. Automated methods for activity recognition of construction workers and equipment: state-of-the-art review. *J. Constr. Eng. Manag.* 146, 03120002. [https://doi.org/10.1061/\(asce\)co.1943-7862.0001843](https://doi.org/10.1061/(asce)co.1943-7862.0001843).
- Snyderman, R., 2012. Personalized health care: from theory to practice. *Biotechnol. J.* 7, 973–979. <https://doi.org/10.1002/biot.201100297>.
- Tang, N., Hu, H., Xu, F., Yeoh, J.K.W., Chua, D.K.H., Hu, Z., 2022. A personalized Human Factors Analysis and Classification System (HFACS) for construction safety management based on context-aware technology. *Enterp. Inf. Syst.* 16, 141–166. <https://doi.org/10.1080/17517575.2021.1878283>.
- Umer, W., Li, H., Lu, W., Szeto, G.P.Y., Wong, A.Y.L., 2018. Development of a tool to monitor static balance of construction workers for proactive fall safety management. *Autom. Constr.* 94, 438–448. <https://doi.org/10.1016/j.autcon.2018.07.024>.
- Xia, N., Zou, P.X.W., Liu, X., Wang, X., Zhu, R., 2018. A hybrid BN-HFACS model for predicting safety performance in construction projects. *Saf. Sci.* 101, 332–343. <https://doi.org/10.1016/j.ssci.2017.09.025>.
- Xu, M., Nie, X., Li, H., Cheng, J.C.P., Mei, Z., 2022. Smart construction sites: a promising approach to improving on-site HSE management performance. *J. Build. Eng.* 49, 104007 <https://doi.org/10.1016/j.job.2022.104007>.
- Yan, X., Li, H., Wang, C., Seo, J.O., Zhang, H., Wang, H., 2017. Development of ergonomic posture recognition technique based on 2D ordinary camera for construction hazard prevention through view-invariant features in 2D skeleton motion. *Adv. Eng. Informatics* 34, 152–163. <https://doi.org/10.1016/j.aei.2017.11.001>.
- Yan, X., Li, H., Zhang, H., Rose, T.M., 2018. Personalized method for self-management of trunk postural ergonomic hazards in construction rebar ironwork. *Adv. Eng. Informatics* 37, 31–41. <https://doi.org/10.1016/j.aei.2018.04.013>.
- Yang, K., Ahn, C.R., 2019. Inferring workplace safety hazards from the spatial patterns of workers' wearable data. *Adv. Eng. Informatics* 41, 100924. <https://doi.org/10.1016/j.aei.2019.100924>.
- Yang, K., Ahn, C.R., Vuran, M.C., Kim, H., 2017. Collective sensing of workers' gait patterns to identify fall hazards in construction. *Autom. Constr.* 82, 166–178. <https://doi.org/10.1016/j.autcon.2017.04.010>.
- Yoon, J., Lee, B., Chun, J., Son, B., Kim, H., 2022. Investigation of the relationship between Ironworker's gait stability and different types of load carrying using wearable sensors. *Adv. Eng. Informatics* 51, 101521. <https://doi.org/10.1016/j.aei.2021.101521>.
- Yu, Y., Guo, H., Ding, Q., Li, H., Skitmore, M., 2017. An experimental study of real-time identification of construction workers' unsafe behaviors. *Autom. Constr.* 82, 193–206. <https://doi.org/10.1016/j.autcon.2017.05.002>.
- Yu, Y., Li, H., Umer, W., Dong, C., Yang, X., Skitmore, M., Wong, A.Y.L., 2019. Automatic biomechanical workload estimation for construction workers by computer vision and smart insoles. *J. Comput. Civ. Eng.* 33, 04019010. [https://doi.org/10.1061/\(asce\)cp.1943-5487.0000827](https://doi.org/10.1061/(asce)cp.1943-5487.0000827).
- Zhang, M., Fang, D., 2013. A cognitive analysis of why Chinese scaffolders do not use safety harnesses in construction. *Constr. Manag. Econ.* 31, 207–222. <https://doi.org/10.1080/01446193.2013.764000>.
- Zhao, J., Obonyo, E., 2020. Convolutional long short-term memory model for recognizing construction workers' postures from wearable inertial measurement units. *Adv. Eng. Informatics* 46, 101177. <https://doi.org/10.1016/j.aei.2020.101177>.